Prabhavit Gupta

Forward Deployed AI Engineer | Production RAG, Agentic Workflows (LangGraph/Pydantic) & LLMOps Evals

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Executive Summary

- **Forward Deployed AI Engineering & Production Delivery:** Client-facing Forward Deployed AI Engineer with 3 years commercial software experience turning experimental AI prototypes into robust, production-grade applications across European tech platforms.
- **Evaluation-Driven LLM & Agentic Architectures:** Deep specialization in agentic state machines (LangGraph, Pydantic AI), advanced RAG retrieval, and quantitative evaluation test harnesses (DeepEval, RAGAS). Systematic builder who establishes baseline metrics to prove that prompt, retrieval, or model alterations measurably move business KPIs.
- **Pragmatic Python Engineering & Consultant Mindset:** Experienced embedding directly with cross-functional stakeholders and engineering leadership to translate ambiguous business challenges into scalable technical roadmaps. Full EU right to work with immediate readiness to relocate to Amsterdam.

Technical Skills & Core Competencies

- **GenAI Frameworks & Agents:** LangGraph, Pydantic AI, LangChain, Model Context Protocol (MCP), Multi-Agent Tool Orchestration, Context Pruning, Human-in-the-Loop Interventions.
- **Production RAG & Vector Search:** Advanced Hybrid Search, Reciprocal Rank Fusion, pgvector (HNSW Indexing), Chunking & Re-ranking Strategies, Embeddings Pipelines.
- **LLMOps, MLOps & Evals:** DeepEval, RAGAS, Langfuse Distributed Tracing, Golden Dataset Benchmarking, CI/CD Automated Evaluation Gates, Drift Detection.
- **Python & Cloud Engineering:** Python (AsyncIO, Typing, PyTest), FastAPI, PostgreSQL, Redis, Docker, Git, CI/CD Automation, AWS/GCP Cloud Deployments.
- **Client Engagement & Consulting:** Technical Scoping, Architecture Reviews, Stakeholder Management, Solution Workshops, Pair-Programming Debugging.

Professional Experience

Forward Deployed AI Engineer & Solutions Consultant • *Independent / AI Product Engineering Consultancy*
2024 – Present • Venice, IT (Remote / Hybrid Europe)

- Embedded with European client teams to architect, deploy, and harden production-grade GenAI applications, moving customer proofs-of-concept into resilient production environments.
- Architected multi-agent workflows using LangGraph and Pydantic AI, coordinating state persistence, deterministic tool access, and dynamic error recovery.
- Built evaluation-driven testing harnesses using DeepEval and RAGAS, establishing precision, recall, and hallucination baselines that systematically validated model updates in CI/CD.
- Standardized model telemetry and observability using Langfuse distributed tracing, monitoring real-time latency profiles, tool execution reliability, and token usage budgets.
- Advised client CTOs and engineering leads on scalable LLM deployment patterns, data governance boundaries, and architectural tradeoffs.

Full-Stack AI Software Engineer • *Tech Automation Systems*
2023 – 2024 • Milan / Remote, IT

- Developed asynchronous microservices in Python (FastAPI) and TypeScript for automated document ingestion, structured entity extraction, and intelligent classification.
- Engineered high-performance vector retrieval layers combining PostgreSQL (pgvector) and Redis semantic caching, reducing duplicate LLM calls by 55% while improving response latencies.
- Authored automated unit and integration test suites using PyTest, implementing rigorous quality controls across Git-based deployment workflows.
- Partnered closely with enterprise clients to translate unstructured domain knowledge into modular, maintainable system architectures.

Key Engineering Projects

- **Evaluation-Driven Multi-Agent Delivery Pipeline (LangGraph + Pydantic AI + RAGAS):** Architected an enterprise agent loop with isolated retrieval and validation nodes, complete with automated regression evaluation gates catching prompt drift.
- **Production Hybrid RAG & Vector Search Engine (FastAPI + pgvector + Langfuse):** High-throughput semantic search service combining dense vector similarity with lexical filtering, profiled with real-time Langfuse

tracing.

Education, Languages & Legal Status

- **Bachelor of Arts in Philosophy & Economics** – Ca' Foscari University of Venice (2018 – 2022)
  Deep grounding in formal logic, decision analysis, epistemology, and complex mathematical modeling.
- **Languages:** English (C1 Full Professional / Fluent), Italian (Working proficiency), Hindi (Native).
- **Work Authorization & Legal Status:** EU Long-Term Resident (Permesso di Soggiorno UE a Tempo Indeterminato, Italy) – Unrestricted legal right to live and work in the Netherlands without visa sponsorship; available immediately for Amsterdam hybrid relocation.